THE UNIVERSITY OF HONG KONG
HKU BUSINESS SCHOOL
2020-2021 2nd Semester Midterm Examination

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Finance: FINA2322
Derivatives
Sub-class E, F, G: Dr Yang Liu

Mar 16, 2021 6:00-7:30 p.m.

There are two parts in the exam.

Part I. 7 Multiple Choice Questions (5 point each, 35 points in total)

Part II. 7 Short Questions (65 points in total)

The total points are 100.

Total Pages of the Exam Paper (including this cover page): 7

- **You have to write down your Name and UID on the Answer Script.**

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A. I acknowledge that University examinations require all students to respect the highest standards of academic integrity. For the examination I am about to take, I make the following pledge:

1. All the work will be my own, and I will not plagiarize from anyone else;

2. I will not obtain or seek to obtain an unfair advantage by communicating or attempting to communicate with any other person during the examination; neither will I give or attempt to give assistance to another student in taking the examination;

B. I understand that students who are suspected of violating this pledge are liable to be referred to the Disciplinary Committee, and maybe subject to disciplinary action such as suspension of studies or expulsion from the University.

Case 1 Hong Kong Gold Mining Limited

Hong Kong Gold Mining Limited (HKGM) is your client and has gold mining business in all over the world. With the huge virus shock to the economy, the gold price has been very volatile and appears to have a downward trend recently. HKGM has come to you for advice in risk management.

First of all, HKGM sells gold in operation. They are considering whether or not to hedge their gold price risk. You explained to them what the advantages of hedging are and also explained reasons not to hedge.

MC Question 1 (5 points)
Which of the following is a not reason that HKGM wants to hedge risk?
A) Monitoring costs
B) Bankruptcy costs
C) Managerial Risk Aversion
D) External financing costs

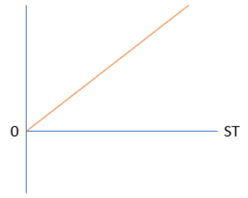
MC Question 2 (5 points)
HKGM has an inherent ___(1)___. They could hedge his position by ___(2)___.

	(1)	(2)
A)	Short position in the gold price.	Long a call on gold.
B)	Long position in the gold price.	Short a collar on gold.
C)	Short position in the gold price.	Short a put on gold.
D)	Long position in the gold price.	Short a forward on gold.

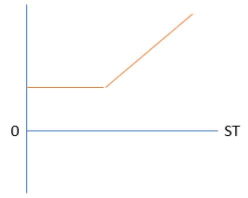
Short Questions 1 (6 points)

HKGM has scheduled to sell gold exactly 9 months from now (1 March). HKGM is considering a strategy that would hedge the downside risk of gold price without affecting the upside position (strategy 1). The strike price chosen in strategy 1 is $1700 per ounce.

(a) What is the unhedged revenue per ounce? Draw the payoff diagram. (3 points)



(b) What is the hedged revenue per ounce? Draw the payoff diagram. (3 points)



HKGM found that strategy 1 appears to be costly and would like to pay a lower cost for the insurance strategy.

MC Question 3 (5 points)

Which of the following insurance strategy is more costly than the original strategy 1?

A) Long Collar with strike prices $1700 and $1800 per ounce

B) Short Forward

C) Long put with strike price of $1800 per ounce

D) Short call with strike price of $1800 per ounce

Your client is excited to learn about the forward contract. He asked about whether he can use the forward contract to predict the gold price in the future.

Short Question 2 (4 points)

Explain whether the forward price can be used as a prediction of the gold price in the future.

No. Forward price is calculated based on risk-free interest rate, expected stock price is calculated based on expected stock return, which should be larger than risk-free interest rate.

HKGM asked about the differences between the long forward contract on gold and a direct investment in gold.

MC Question 4 (5 points)

You have some thoughts when you explain to HKGM about forwards.

1. When you take a short position in the forward contract, the initial value of the contract is zero.
2. When you buy a forward, you expect to earn compensation for the time value of money.
3. When you buy a forward, you expect to earn compensation for risk.
4. When there are transaction costs, the no arbitrage forward price becomes a range instead of a number.

How many of the above statements are true?

A) 1

B) 2

C) 3

D) 4

Short Questions 3 (17 points)

HKGM has a trading department to speculate on gold price. The department expects gold price to be very volatile in the coming 1 year, i.e. the gold price might rise a lot or decreases a lot.

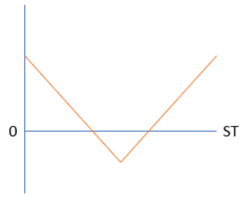
Eventually, HKGM decides to profit from the expected situation of gold price and is considering using straddle to help capture the profit (strategy 2).

The following options expire in 1 year and the 1-year interest rate is 4% p.a continuously compounded. The gold price for today (March 1) is $1700 per ounce. Below is the relevant call and put premiums:

Strike	Call premium	Put Premium
1600	285.03	122.29
1700	233.81	167.15
1800	190.08	219.5

(a) Find out the cost of strategy 2. (2 points)
233.81+167.15 = 400.96

(b) Draw the profit diagram of strategy 2. (4 points)



(c) From the profit diagram, what is the potential risk of using strategy 2? (2 points)
When the gold price stayed at $1700.

(d) As the counterparty of HKGM for strategy 2, you face some risks and want to hedge them using 1600-1800 strangle. Find the cost of strangle. (3 points)
122.29+190.08=312.37

(e) What is the range of gold prices such that the profit of your combined position is positive? (6 points)
FV of cost of combined strategy = (312.37-400.96)*e^(0.04) = -92.205
Range: 1700 - 92.21 < S_T < 1700+92.21 → 1607.79 < S_T < 1792.21

Case 2 Derivatives Asset Management

You are a fund manager in Derivatives Asset Management Fund (2322.HKU). You manage a portfolio consists of stocks and derivatives contracts. You profit mainly from arbitrage activities from mispricing and speculating on the movements and volatility of the market.

On a regular day, you will project your price expectation on the market. The current Hang Seng Index level is 28500. The 6-month risk-free interest rate is 0.1% p.a., continuously compounded. The dividend yield of HSI is expected to be 3.5% p.a., continuously compounded for the coming 6 months. Below is your expectation of HSI level 6 months later:

Probability	Hang Seng Index level
0.1	27000
0.3	28500
0.4	29000
0.2	30000

MC Question 5 (5 points)

Based on your price expectation, which of the following strategy is least likely to capture a profit?

A) Long Call
B) Bull Spread
C) Long Forward
D) Long Put

In order to speculate in the market, you are considering the following strategies:

Strategy 1: Long Call option with a strike price of 28500, short put option with a strike price of 27000
Strategy 2: Long Call option with a strike price of 27000, short call option with a strike price of 30000
Strategy 3: Long Call and Long Put with a strike price of 28500

All of the options used are with 6-month maturity. Below is the relevant call and put premiums:

Strike	Call premium	Put Premium
27000	2112.39	1093.31
28500	1363.65	1843.82
30000	831.38	2810.8

Short Questions 4 (13 points)

(a) What is the expected profit of strategy 1? (4 points)
Profit = 0.4*500+0.2*1500+0-(1363.65-1093.31)*e^(0.001/2) = 229.52

(b) What is the expected profit of strategy 2? (4 points)
Profit = 0.3*1500+0.4*2000+0.2*3000+0-(2112.39-831.38)*e^(0.001/2) = 568.35

(c) What is the expected profit of strategy 3? (4 points)
Profit = 0.4*500+0.2*1500+0.1*1500-(1363.65-1843.82)*e^(0.001/2) = -2559.07

(d) Which strategy allow you to capture the highest profit? (1 points)
Strategy 2

Short Questions 5

Suppose you try to construct a synthetic forward contract with options such that you buy the call option with a strike of 28500 and short a put option with a strike of 28500. The maturity of both options is 6-month. What is the profit of the synthetic forward contract? Express the profit in terms of S_T, there is no need to consider the probability.
Profit = Max(S_T – 28500, 0) – Max (28500 – S_T, 0) – (1363.65-1843.82)*e^(0.001/2) = S_T – 28019.59

MC Question 6 (5 points)
Based on your answer in short question 5, what arbitrage can you take if the 6-month forward price is 28500?

A) Long call, short put, short forward
B) Long call, short put, long forward
C) Short call, long put, short forward
D) Short call, long put, long forward

MC Question 7 (5 points)

Suppose the 6-month forward contract is correctly priced instead of 28500 and you have taken a short forward position. If 2 months later, the value of HSI changed to 28000 and the 4-month interest rate is 0.08% p.a. continuously compounded. The dividend yield of HSI remained the same at 3.5% p.a., continuously compounded for the coming 4 months. What is the value of your short forward position?
A) Zero
B) Positive
C) Negative
D) Unknown

You observed the forward price in the market is 28000 for a contract with 1 year to maturity. The 1-year risk-free interest rate is 0.2% p.a., continuously compounded. The dividend yield is still 3.5% p.a. continuously compounded. You planned to capture an arbitrage profit. Assume the size of a forward contract is 1 times the index level. Assume there are no transaction costs. Note that the current value of HSI is still 28500.

Short Questions 6 (8 points)

(a) What is the fair 1-year forward price? (3 points)
28500*e^(0.002-0.035) = 27574.85

(b) How would you capture the arbitrage profit and what is the profit? (5 points)

Hint: You may use the following table to help you get the answer:

Strategy	Time = 0	Time = 1
Short Forward	0	28000 - S_T
Long Stock	-28500*e^(-0.035) =-27519.75	S_T
Borrow	27519.75	-27519.75*e^(0.002) =-27574.85
Total	0	425.15

Short Questions 7 (13 points)

Suppose there are transaction costs. The lending rate and borrowing rate for 1 year are 0.15% p.a. and 0.25% p.a. respectively, continuously compounded. When you buy or sell the index, there is a transaction cost of 1% at t=0. There is no transaction cost for the stock 1-year later. There is also a fixed transaction cost for the forward contract of $25 per contract at t=0.

(a) If you arbitrage as in SQ6 with a forward price of 28000, what is the profit? (5 points)

Strategy	Time = 0	Time = 1
Short Forward	-25	28000 - S_T
Long Stock	-28500*e^(-0.035)*1.01 =-27794.95	S_T
Borrow	27819.95	-27819.95*e^(0.0025) =-27889.59
Total	0	110.41

(b) What is the no-arbitrage upper bound for the forward contract? (3 points)

S_T =< 27889.59

(c) What is the no-arbitrage lower bound for the forward contract? (5 points)

Strategy	Time = 0	Time = 1
Long Forward	-25	S_T - F
Short Stock	28500*e^(-0.035)*0.99 =27244.56	-S_T
Lend	-27219.56	27219.56*e^(0.0015) =27260.42
Total	0	27260.42 - F

Set 27260.42 – F =< 0 → F >= 27260.42